**H. CARL HAYWOOD,** *Vanderbilt University, U.S.A.*

# *A Mediational Teaching Style*

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The mediational teaching style is a direct derivative of Feuerstein's Theory of Structural Cognitive Modifiability. It is an application by teachers in classrooms of principles of adult-child interactions, referred to as "mediated learning experiences" (MLE), that are thought to be essential for the adequate cognitive development of children. Such interactions typically take place between children and older members of their families such as parents, grandparents, older siblings, and other caretakers. These interactions have the function of "mediating" the generalized meaning of the world to the children; that is, they help children to understand that events, objects, and persons have meaning beyond themselves, that the universe has predictable structure, that understanding that structure helps one to know what to do in a wide variety of future situations, that it is possible to make explanatory rules that help one to organize observations, and that it is essential to test the applicability of such rules in a wide variety of circumstances. The intermediate-range goal of MLE is to acquire the fundamental cognitive functions that underlie the ability to learn effectively across many and varied content fields.

## The Role of MLE

The role of parents in their children's cognitive development represents a major difference between the developmental theories of Piaget and those of Feuerstein (Arbitman-Smith, Haywood, & Bransford, 1984). Piaget gave scant attention to the role of child-rearing agents such as parents, grandparents, and teachers (Haywood & Wachs, 1981), but in Feuerstein's view such persons play a central and critical role (Feuerstein & Rand, 1974). In fact, Feuerstein has maintained that adequate cognitive development is not possible without some degree of MLE provided by parents and/or other caregivers. He has presented a schematic diagram showing his idea of the "proximal" and "distal" etiologic conditions that lead to either adequate or inadequate development of the most fundamental cognitive functions (Feuerstein, Rand, Hoffman & Miller, 1980). According to this scheme, the various conditions that have traditionally been thought of as "causes" of inadequate cognitive development, such as poverty, neurological impairment, emotional disturbance in child or parents, and low educational levels of the parents, are presented instead as correlates or "distal etiologic conditions." That means that even though inadequate cognitive development is found more frequently when those conditions are present than when they are not present, the conditions themselves do not cause inadequate cognitive development.

Instead, the "proximal" (immediate) source of inadequate cognitive development is lack of adequate MLE. The correlation arises because those distal etiologic conditions, or correlates, are often, but not necessarily, associated with inadequate MLE. Feuerstein et al. (1980) have contended that when MLE is sufficient for individual needs of particular children, adequate cognitive development will be the result, and that when MLE is not sufficient, inadequate cognitive development, and the syndrome of "cultural deprivation" (see discussion below) will be the result. Thus, Feuerstein et al. (1980) have suggested that adequate cognitive development can occur in spite of such "distal" conditions as poverty, neurological impairment, mental retardation, emotional disturbance, and low educational levels, when there is adequate mediation of the most fundamental cognitive functions to the children by more competent older persons. Further, they have suggested that inadequate cognitive development may occur in spite of favorable circumstances with respect to these "distal" conditions, that is, in favorable economic circumstances and in the absense of neurological impairment, mental retardation, emotional disturbance, and low educational levels, when MLE is not sufficient to meet the individual developmental needs of particular children.

The condition of poverty is a good example of this correlated relationship. Haywood and Stedman (1969) have noted that over 80 percent of all mildly mentally retarded children come from "poor" families, so one might be tempted to conclude that poverty causes mild mental retardation. At the same time, these authors also noted that only 10 percent of children from poor families are ever identified as mentally retarded. If 90 percent of families in which the condition of poverty is present produce children whose intellectual functioning is above the cutting point for the diagnosis of mild mental retardation, then it would be quite illogical to argue that poverty causes mental retardation. It is also true that 10 percent of children from poor families do get identified as mentally retarded, and that is more than three times the prevalence of mental retardation found in the population as a whole. Thus, while poverty does not cause mental retardation, it is clear that the social circumstance of poverty may contain some elements (positive or negative) that bring about a correlation (in this case, a greater likelihood of occurrence) with mental retardation. The position of Feuerstein et al. (1980) is that the correlation is brought about by the fact that adequate mediated learning experience is less likely to be provided, for a variety of reasons, when families are economically and socially deprived than when they are economically and socially secure.

It is clear, then, that "adequate" MLE is a critical condition for adequate cognitive development, according to Feuerstein's position. That being the case, one might well ask "How much MLE is enough?" The answer is that it depends on the individual needs of the child. Haywood (1989; Haywood & Switzky, 1992) has made a sharp distinction between "intelligence" (native ability, relatively fixed and constant, largely genetically determined) and "cognitive pro-

cesses" (modes of logical thought, must be learned, can be taught, highly modifiable through teaching), and this distinction can be helpful in understanding this relationship. Feuerstein et al. (1980) have suggested two ways in which basic cognitive processes are acquired: (a) through "direct exposure" learning, that is, they can be learned without external mediation through children's direct successive encounters with environmental events; (b) mediated learning experience, that is, through the mediation ( by adults) of the generalized meaning of those events. Every child, no matter how "intelligent," must acquire the basic cognitive functions in order to think logically, perceive the world in structured, orderly, and reasonable ways, know how to learn, and apply his/her intelligence to new learning and problem solving situations without further need of mediation. Children who are genetically (or otherwise) destined to have relatively high intelligence may acquire a larger proportion of their basic cognitive functions through direct exposure learning than do children of initially lower intelligence, and further may require less frequent, less repititious, and less intense MLE in order to acquire their basic cognitive functions. On the other hand, children who have relatively lower intelligence, or who have special handicapping conditions such as limited vision or hearing, may learn less through direct exposure to environmental events, thereby demonstrating a greater need for MLE, and further may require more frequent, more repetitious, and more intense MLE in order to acquire their basic cognitive functions. The developmental question with such children is not whether they can acquire important thinking modes and processes, but *how much mediated learning experience,* of what kinds, over what period of time, will be necessary in order for them to acquire these processes. To the extent that such MLE is available, the cognitive development of these children will be more nearly adequate than it would have been with less MLE. Failure of either group of children to get adequate MLE will result in inadequate cognitive development, the "syndrome of cultural deprivation," a greater discrepancy between their performance and their potential than would have been necessary with more MLE, and less effective learning of both academic and social content than would be potentially possible for them.

Feuerstein has referred often to the "syndrome of cultural deprivation." One must be careful to distinguish his use of this term from the widely understood definition of the 1960s. Between 1960 and the early 1970s, American developmental psychology and early education were characterized by enormous optimism about the possibility of raising intelligence through education, especially early education, and by the notion that some cultures and subcultures were inherently "depriving" with respect to some of the necessary component of "good" childbearing environments. Children who were reared in such cultures and subcultures were seen as "culturally deprived," meaning that their cultures had deprived them of essential developmental experiences, and that it would be somehow necessary to "compensate" for those lacks. Feuerstein has maintained that such a notion of cultural deprevation is false and misses the essential point. He asserts

that every culture and subculture that has been around long enough to warrant the name contains all the social elements that are essential for adequate cognitive development. In the vast majority of families, these elements are transmitted to the children through a broad set of processes that he calls "intergenerational cultural transfer" — the passing of a culture's essential elements from one generation to the next — that relies heavily on MLE as its chief tool. It is when environmental events interrupt that transfer, that is, when children are somehow denied the benefits of their own culture, that the children can be regarded as "culturally deprived." Thus, culturally deprived children, rather than having been denied essential developmental events by their cultures, have instead failed to receive those events even though all essential elements are present in their cultures. It is failure to get adequate mediation of culture-characteristic ways of perceiving and organizing the world, thinking about thinking, and defining and solving problems that constitutes the syndrome of cultural deprivation. The optimistic side of this view is that, since all the essential elements are present within one's culture, deficiencies in their transmission can be made up by subsequent mediation. That often becomes the role of teachers.

## Summary of the Role of MLE

It is possible to summarize the important points about MLE from a combination of Feuerstein's Theory of Structural Cognitive Modifiability and Haywood's transactional perspective on the nature and development of intelligence. (Haywood & Switzky, 1992; Haywood, Tzuriel & Vaught, 1992.) The following list contains the essential points.

1. Intelligence itself is relatively constant, and efforts to change it by education yield only modest increases as a result of substantial investment.
2. Intelligence alone (defined as "native ability") is not a sufficient condition to provide for effective perception, thinking, learning, and problem solving.
3. There is a finite number of fundamental processes of thinking that, combined with certain affective, motivational, and attitudinal factors and particular habits of intellectual work, constitute "basic cognitive functions" and are necessary for effective systematic perceiving, thinking, learning and problem solving.
4. The basic cognitive functions must be acquired (i.e., are not genetically "given"), and they are acquired through learning.
5. Cognitive learning is of two kinds: direct exposure to environmental events, and mediated learning experience (MLE).
6. Some MLE is necessary for all children, but the amount, quality, intensity, frequency, and duration of what is needed for adequate cognitive development will vary as a function of individual differences in children (e.g., genetic intelligence levels, sensory integrity, emotional stability, environmental support).
7. When MLE is inadequate to children's individual developmental needs, the result is inadequate cognitive development, the "syn-

drome of cultural deprivation," and relatively ineffective academic and social learning.

8. When MLE is adequate to children's individual developmental needs, the retarding effects on academic and social learning of such conditions as poverty, mental retardation, emotional disturbance, and low parental education levels may be significantly offset. The result is a higher level of cognitive development and relatively more effective academic and social learning.

9. Providing MLE is an essential role of parents, grandparents, and older siblings in a process of intergenerational cultural transfer. This process is repeated in every culture, and all cultures have within them the essential elements of adequate cognitive development of their children. Failure of transmission of culture-characteristic modes of thinking constitutes the syndrome of cultural deprivation.

10. When some aspects of cognitive development have not been stimulated adequately through MLE, it is possible to mediate those aspects at later ages through carefully constructed teaching. In other words, opportunities that might have been missed by parents can be picked up by other teachers.

11. The mediational processes can be described and replicated, and these processes constitute an identifiable and important style of teaching: the mediational teaching style.

## Qualities of Mediated Interactions

According to Arbitman-Smith, Haywood, and Bransford (1984), "Mediating the learning experiences of children...includes, but is not limited to, such functions as stimulus selection (helping children to reduce the number and complexity of available stimuli by attending selectively), focusing on relevant aspects of a stimulus complex, repeating exposure to important stimuli, perceiving and understanding similarities and differences, sequential relationships, dimensionality, antecendents and consequences, commonalities in experience, and such operations as comparing, categorizing, relating past, present, and future, and grasping the concept of generalizability of experience to new situations". In addition to those rather broad functions, mediators:

1. supply the information that may be needed to learn relationships or to find solutions;
2. ask questions, i.e., elicit rather than give answers;
3. guide children's learning by arranging and directing sequences to experiences in a developmental fashion;
4. bring about induction of explanatory rules by calling attention to similarities among isolated events;
5. guide deduction of applications of rules;
6. build the confidence of the children by communicating belief in their competence as thinkers, and,
7. maintain a *metacognitive* emphasis, i.e., focus attention on the children's own thinking processes and encourage them to do so.

In doing these things, mediators make use of several quite specialized strategies, including *process question-*

*ing, bridging, challenging (requiring justification), teaching about rules, and emphasizing order, predictability, system, sequence, and strategies.*

## Criteria of Mediated Learning Experience

A teaching style is concerned not with only *what* one teaches but primarily with *how* one teaches it. Every interaction between an adult and a child has some potential for being a mediated interaction, that is, for being generalized beyond the content of the immediate situation. Whether or not a given interaction will be useful in promoting the cognitive development of children will depend upon the quality of that interaction. Feuerstein has suggested a number of criteria of mediated learning interactions. From this list I present the first six criteria and suggest that they are the most important. Every interaction can be examined to determine to what extent it reflects each of the six criteria. Awareness of these criteria will encourage parents and other teachers to try consciously and deliberately to construct their interactions with children in such ways as to reflect these criteria.

**Intentionality** - In mediated interactions the mediator intends to use the interaction to produce cognitive change in the child. Content situations (e.g., need to know color names, wish to understand how birds fly) thus become vehicles for the teaching and learning of concepts and principles of thought, which produces cognitive change in the child.

**Transcendence** - The mediator tries to produce structural cognitive changes in the child, i.e., changes that transcend the immediate situation. An event could be seen as only an isolated event, but a mediational teacher will give such an event transcendent (generalized) meaning by attempting to relate the event to previous and even future events of a similar nature, and thus to extract a generality (explanatory rule).

**Communication of meaning and purpose** - mediators do not have cognitive or strategic secrets from children. It is important to let the children know why one is doing any particular activity, i.e., in order to produce structural cognitive changes in the children. Mediators communicate both the immediate (content) meaning of events and their generalized relationship to other events (e.g., "Why do you think it is important for us to do this? Yes, so we will have a plan, and know what to do as we go along!").

**Mediation of a feeling of competence** - Children's feelings about their own competence as learners are extremely important. Mediators communicate about this in two specific ways. The first way is to reward appropriate responses, especially process-oriented responses, with acceptance, acknowledgment, praise. The second is to be certain that the children understand exactly what aspects of their behavior were good and should be repeated. Thus, a mediator does not stop at saying "Good!" when a child has done well, but might say something such as "Good! You made a plan, so now you know what to do as you go along."

**Regulation of behavior** - Children, especially young children, often require

some regulation of their behavior in order to demonstrate their cognitive competence. A very common source of errors in intellectual work is impulsive responding, i.e., giving answers before one has had time to examine the questions and the possible solutions. A good mediator helps children to inhibit their impulsive responding and thus to improve the quality of their responses. Another manifestation of need for regulation of behavior is inability or unwillingness to respond even when the mediator knows that responses are available to the children. Good mediators help children to "unblock" available responses, to be willing to give answers, by creating an affectively safe and expectant environment.

**Shared participation** - Mediators convey the attitude that they and the children are engaged in a shared quest for structural cognitive change in the children. Each has an identifiable and separate role, but each is a participant and shares in a "We're in this together and you can count on me to do my part" manner. The interaction, then, is not one-way, and while it is directive it is neither authoritarian nor patronizing.

Mediated interactions differ demonstrably from nonmediated interactions in several ways. First, of course, mediated interactions are characterized by the criteria discussed in the preceding paragraphs. Second, mediated interactions are strongly process-oriented; that is, the dialogue between child and adult is focused on generalizable processes of thinking and learning rather than on answers to immediate problems in immediate situations. Third, mediated interactions display mediators' confidence in children's ability to learn and apply appropriate thinking modes and strategies. In contrast to many kinds of nonmediated interactions, mediated ones constitute true dialogue, that is, a "two-way street" in which information is sought by each participant from the other and the unique role of each participant is recognized. These distinctions are both conceptual and empirical. At the empirical level, Dale and Cole (1988) carried out a systematic comparison of CCYC (Haywood, Brooks & Burns, 1992) and DISTAR (Becker, Engelmann, & Thomas, 1975; Becker, 1977), the latter a content-oriented curriculum whose teaching style is substantially behavioristic. The two curricula differ markedly in concept and approach, but these authors asked to what extent they differ in actual implementation; i.e., they wanted to know to what extent observers could distinguish the two approaches on observational criteria. First, they developed 14 categories of teacher and child behavior on which the programs were expected theoretically to differ. Observers were trained, and made their observations according to strictly prescribed protocols. There were significant differences on 9 of their 14 categories, all in the expected directions. DISTAR exceeded CCYC in frequency of: eliciting verbal imitation, eliciting unison responding, limited response questions, "labelling" questions, and immediate correction (reinforcement). CCYC exceeded DISTAR in frequency of: process presentation by the teachers, process generalization, process questions, and open-ended observation questions. The researchers commented that "It is

certainly true that the difference between the programs is clear virtually immediately on entering the classrooms." Thus there is evidence that teachers who have been well trained in the mediational teaching style do indeed behave in their classrooms in ways that are conceptually consistent with the requirements of that teaching style, and that their classroom behavior is visibly different from that of teachers trained in a different teaching style.

### Unique Character of Teacher-Mediated Interactions

What teachers do by way of mediation of cognitive functions in classrooms is indeed a way of compensating for inadequate MLE in home situations, but the two are not the same. That is to say, when mediation takes place in home and family settings, and is done by primary child-rearing agents, it is of a different quality than when it is done in classrooms by teachers. Some of the differences should make it possible to realize more efficient gain from the classroom procedures.

1. Since the efforts of teachers occur later than do the primary efforts of parents, the children are more competent, presenting a developmentally more complex set of abilities, habits, attitudes, and expectations as well as a richer accumulation of experience. This important difference means that teachers may move at a more rapid pace, that they can draw on a richer and more varied store of familiar experience in the children (to which they can bridge cognitive concepts), and that the children themselves can be counted on to participate more actively and confidently.
2. Family mediators rely upon naturally occuring situations as opportunities to mediate, but classroom teachers, having many children to whom they must mediate at once, typically construct their own opportunities to mediate. The contrived situations used by classroom teachers allow the teachers to control the parameters of mediational situations, to set clear mediational goals, and to sequence the children's cognitive learning in developmental ways.
3. Family mediators, relying on naturally occuring situations, use relatively "concrete" situations, and must make the effort to extend the meaning of these in cognitively generalizable ways. Classroom teachers, using contrived and structured situations, go more directly to representational thought; in fact, their "bridging" is done frequently to other "concrete" or real-life situations when the same cognitive functions are required or useful.
4. Classroom teachers make use of the social situation of having many children of similar developmental age at once, while in families there is rarely more than one child of a given developmental level. The presence of other children can be a powerful advantage in the mediational process, since the children sometimes understand other children's examples and experience more readily than they understand the experience of adults, and there is a certain amount of social facilitation in the learning process. Of

course, these same phenomena can work to the disadvantage of classroom teachers to the extent that the social interactions are negative or destructive.

5. In general, classroom mediation is more structured, more clearly teacher directed, more focused upon specific cognitive goals, and more concentrated (in the sense that similar cognitive functions are being emphasized throughout the day in a cognitive classroom) than is true of family mediation.

## The Most Useful Mediating Mechanisms

I have suggested earlier five mechanisms of mediational teaching that are used frequently and that are thought to be especially useful in helping children in classroom settings to acquire developmentally important thinking processes. These mechanisms are discussed here.

**Process questioning** - This is the single most frequently used mechanism in a cognitive classroom. Teachers ask many questions, and their questions are very often of a process nature. This means that they ask "how" questions: "Yes, but how did you know?" "How else could you do that?" "What must you do first, and how can you find out what to do next?" This mechanism is extremely important as a *metacognitive* tool, i.e., it helps to focus children's attention on their own thinking processes and encourages them to engage in similar "small converstations" with themselves.

**Bridging** - This is the process whereby teachers encourage children to think of different applications of the thinking processes and strategies they have been discussing. It is *cognitive functions* that are bridged. Content may also be bridged, but in a cognitive curriculum the bridging of cognitive functions is more important. Thus, if one is teaching "comparing on multiple dimensions," a teacher might ask the children to nominate other situations (than the ones introduced in the lesson) when it is important or helpful to compare on more than a single dimension. It is probably true that the learning of a concept or principle of thought does not take place merely in the verbalizing of the principle (although there is evidence that that helps!) but in the children's successive attempts to apply the principle in a variety of different situations and applications. Because we as human beings are capable of representational thought, actual trying out of thinking principles in varied real situations is not completely necessary; we can "try out" applications symbolically, by imagining them and by discussing them. The following principles should govern the use of bridging as a mediating mechanism:

1. "Bridges" should be *elicited* from the children, not told to them (although some "pump priming" is often necessary).
2. Bridges should be to events and circumstances that are familiar to the children.
3. Bridging examples should be simple and straightforward, not complex and logically tortured.
4. Bridging examples should be elicited in several domains of experience, especially other school contexts, home situations, and peer group interactions.

**Challenging, or requiring justification** - At first glance, this mechanism might seem to have the potential for decelerating participatory behavior in children. Good mediational teachers establish the habit of challenging both correct and incorrect responses. Challenging must be accompanied by the rule of accepting as much as possible of children's responses (the "Yes, but..." mechanism). One might say, for example: "Yes, you are right, it could be that way. You could also look at it aother way, and maybe find an even better answer." Challenging a correct answer conditions the child against the expectation that a challenge by the teacher means that their answers are wrong. A "correct" challenge might be: "Yes, that's right. How did you know that should be the answer? Why is it better than this one? What would be wrong with this one? Could you show me/tell me how you thought about that and found the right one?"

**Teaching about rules** - This is a critical aspect of generalizability. The idea, of course, is that if one can make an applicable rule one will know what to do in future similar situations. The children themselves should be asked often to generalize: "Can we make a rule about how to do this kind of problem? If redbirds have feathers, bluebirds have feathers, eagles have feathers, grackles have feathers, and sparrows have feathers, can you think of a rule about birds? Does it apply to hawks? To penguins?" In addition to such rule making, mediational teachers should ask the children to comment often on the general utility of rules: "Would it help us to have a rule here? How? How could we make one? Who can tell me two different times when we need to have rules in order to know what to do?"

**Emphasizing order, predictability, system, sequence and strategies** - This one is quite important but perhaps a bit more difficult to define in operational terms. The general attitide of mediational teachers should be that there is order in the universe, that events are predictable if one understands the rules and has enough information, that events, objects, and even persons are arranged in systems, that the order in which one looks at or does things is important, and that problem solving is often primarily a question of finding the most effective strategy. These attitudes are mediated throughout a CogEd day, beginning with planning time and ending with summary time, both of which serve to emphasize the orderliness and predictability of the universe as well as the satisfaction of placing events in agreeable sequence. Within any interaction during the day, these elements can be injected into a discussion on almost any topic.

There are as many specific ways to mediate cognitive functions to children as there are good mediators. In fact, good mediators use their own personalities and the feedback they get from the children to regulate their behavior and to select mediational strategies, rather than memorizing particular mediational sequences and using those over and over with children.

## CORRESPONDENCE

*H. Carl Haywood, Cognitive Education Research Group, Box 9, Peabody Station, Vanderbilt University, Nashville, TN 37203 U.S.A.*

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